



UNIVERSITY OF LONDON

BSc Computer Science

CM2025 Cyber Security

Midterm coursework assignment

Introduction

This midterm coursework assesses Topics 1–5 of the CM2025 Cyber Security module. These include: Identification and classification of cyber threats; Cybersecurity frameworks, standards, legal, ethical and professional considerations; Network and OS security; Application security principles, including testing and debugging; and Cryptography in practice. You must complete both parts (A and B). The assessment is worth 50% of your final module mark. Submit your work as a single PDF report of no more than 2,500 words, excluding code. All references must be cited in either Harvard or IEEE style.

PART A – Research and Analysis (40%)

Task A1: Recent Cyber Threats (20%)

In this task, you are required to research and analyse **FIVE** significant cyber threats that have emerged or evolved since January 2023. For each threat, provide its name and classification (for example, ransomware, phishing campaign, APT, zero-day, or insider threat). Describe how the threat operates, including its propagation mechanism or exploit technique, and indicate which sectors or systems it targets. Discuss possible mitigation and response strategies, referencing practical frameworks, tools, and best practices. All information should be drawn from reliable sources such as CISA, NCSC, MITRE ATT&CK, or cybersecurity industry reports. Each description should not exceed 300 words.

Task A2: Ethical and Framework-based Evaluation (20%)

Select one of the threats you identified in Task A1 and assess it using a cybersecurity framework you are familiar with, such as NIST CSF or ISO 27001. Explain how the threat develops from start to finish and describe its impact on an organisation. Finally, recommend practical actions and policies that could help prevent this type of threat in future.

PART B – Applied Security Practice (60%)

Task B1: Network and OS Security (15%)

Identify and discuss **THREE** common vulnerabilities affecting network or operating system layers. Explain appropriate hardening and protection techniques such as the use of firewalls, intrusion detection and prevention systems, and access control. Include a short discussion of how these defences can be tested and verified using practical tools such as OWASP ZAP or Wireshark.

Task B2: Secure Coding and Application Testing (15%)

Develop or review a simple Python or JavaScript program that contains **TWO** deliberate security flaws. Identify the vulnerabilities, explain their implications, and then demonstrate how to fix them using secure coding practices. Provide evidence of testing and debugging to confirm that the flaws have been mitigated. You should include both the flawed and corrected versions of the code, accompanied by a brief explanation.

Task B3: Cryptography in Practice (30%)

Write a Python program that performs simple AES encryption and decryption of a text message using a shared key. Show the original text, the encrypted version, and the decrypted result. Explain in your own words how the encryption works. Then describe two ways attackers could try to weaken or break encryption and how to defend against each method. Finally, compare symmetric encryption and public key encryption, explaining their key differences, strengths, and when each one should be used.

Submission Format

All submissions must be presented as a single PDF document that includes any relevant code, explanations, and screenshots either within the main body or as an appendix. The total length should not exceed 2,500 words. Your report must include a cover page containing your full name, student ID, module title, and coursework title. All external references must be cited in Harvard or IEEE style. The submission deadline is Week 13, as indicated in the module guide.

Marking Scheme

Section	Description	Weight
Part A1	Research on recent cyber threats	20%
Part A2	Framework and ethics evaluation	20%
Part B1	Network/OS security analysis	15%
Part B2	Secure coding and testing	15%
Part B3	Cryptography implementation and analysis	30%
Total		100% (50% of final module mark)

[END OF COURSEWORK ASSIGNMENT]